Console-based Application with File Storage and OOP Concepts

Bank Management System using Python

Overview of the Project

- A console-based Python program for bank account management.
- Supports account creation, viewing, deposit, withdrawal.
- Uses OOP and dictionary for account storage.
- Extended version includes file-based persistence with JSON.

OOP Concepts Used

- Classes and Objects: BankAccount, BankSystem
- Encapsulation: Each account manages its own data.
- Constructor (__init__): Initializes account details.
- Instance Methods: deposit(), withdraw(), display()

Data Storage and Flow

- Dictionary to store accounts in memory (account_number -> object).
- Upgraded version uses JSON file (accounts.json) for persistence.
- Data loaded on start and saved on changes (deposit/withdraw).

User Interaction and Logic

- Input and Output via console.
- Menu-driven program using while-loop.
- Error handling with try-except for invalid inputs.
- Each menu action handled by a separate function.

Class Diagram (Conceptual)

- BankSystem:
- - Manages multiple BankAccount objects.
- - Handles user input and file operations.
- BankAccount:
- - Holds account_number, name, balance.
- - Methods for deposit, withdraw, and display.

File-Based Version Features

- Accounts saved in 'accounts.json'.
- save_accounts() and load_accounts() methods added.
- to_dict() and from_dict() handle object serialization.
- Data remains safe even after program exit.

Possible Enhancements

- Password protection for accounts.
- GUI using Tkinter or PyQt.
- Web version using Flask or Django.
- Add transaction history or interest calculation.